

# Helix OTA — Continuation

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## 1. Current state

| Field | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| HEAD  | <p>constitution e60cbde adopted (bb3b4189+634fcb50) + CONTINUATION Rev13 (b346323e) + A/B/C stream evidence (de71a634), all pushed 4/4 FF §11.4.113. THIS batch (pending): W1 OpenDesign ground-truth + W2/W3 §11.4.170 host-render harnesses on both frontends (independent-review GO each) + CONTINUATION Rev 14. Prior test-coverage HEAD 21338527.</p> <p><b>OPENDESIGN ADOPTION (2026-07-09)</b> — autonomous loop (§11.4.126). Both operator decisions RESOLVED (keep+improve both frontends; OpenDesign=nexu-io/open-design). §11.4.170 host-render harnesses now live on clients/ota-manager + dashboard (both proven light/dark or light+honest-gap, self-validated). NEXT: feature/ opendesign-adoption — author design-systems/helix-ota/ brand tokens, vendor into both frontends, wire ota-manager theme-toggle DOM class + dashboard dark mode, prove via harnesses. Prior: <b>TEST-COVERAGE PROGRAM</b> (2026-06-23) Phase 0 DONE + Phase 1 mostly done (docs/research/ MASTER_TEST_COVERAGE_PLAN_20260623.md).</p> |
| Phase |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Field         | Value                                                                                                                                                                                                                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | Cuttlefish Tier-2 A/B VERIFIED (F112/F55).                                                                                                                                                                                                                                                                                |
| Terminal goal | Fully validated Helix OTA control plane driving real Android A/B updates end-to-end (protocol round-trip → payload apply → slot switch → rollback) on emulated + physical targets — <b>MET for the emulated Cuttlefish A/B path (F112/F55 VERIFIED); RK3588 stays control-plane-only by operator decision (§11.4.133)</b> |

**Latest session (2026-07-09) — Constitution e60cbde adoption + UI/UX/OpenDesign audit**

Operator directives this session: (1) fetch+pull latest constitution submodule + nested deps, adopt all new tech/rules ASAP; (2) continue the endless autonomous loop with 3-4 parallel subagents on real evidence, no bluff; (3) answer whether Helix OTA has OpenDesign-refined production UI on all platforms.

Landed + pushed 4/4 FF (no force §11.4.113), commits bb3b4189 (semgrep meta-test deletion, split off by a `git add pathspec fatal`) + 634fcb50 (main batch, 29 files +3485/-110): - **Constitution** fc4e9b8→e60cbde (59 commits); nested engines pulled (token\_optimizer @c0591b5, session\_orchestrator @6961c99). Parent pointer bumped. - **Carriers** CLAUDE/AGENTS/GEMINI (.md+HTML+PDF §11.4.65): appended §11.4.167-174, 176-186, §12.12 (pure append, 0 deletions). - **§11.4.166 REPEALED reconciliation** (§11.4.120, not fake-pass): removed CM-SEMGREP-WIRED gate + CM-COVENANT-114-166 propagation + `_semgrep_scan_check` in `commit_all.sh` + the semgrep meta-test; **added** propagation gates 167-174/176-186/§12.12 to `pre_build`. - **§11.4.109 anti-forgetting** wired: `.claude/settings.json` PreToolUse guard-hook (blocks force-push/sudo/host-power/raw-emulator at the tool boundary) + `docs/AGENT_GUARDRAILS.md` (preamble+checklist) + renamed guard meta-test (26/26 bluff-proof). - **Fixes:** `coverage_extra_test.go` `execStubMu mutex` → `ETXTBSY` `fork+exec fd race` (golang/go#22315; 20/20 -count, 10× -race clean); `guard_benchmark_baseline.sh` GUARD-SKIP when benchstat absent (regression 9/9); `export_docs.sh` +weasyprint PDF engine (pandoc 3.10 EXIT-0, genuine PDF 1.7 — a haiku delta-reviewer’s NO-GO on this was **refuted by captured same-conditions evidence** per §11.4.7, see qa-

results/delta\_review\_20260709/refutation.md). - **UI/UX audit answer (docs/research/ui\_ux\_opendesign\_audit\_20260709/FINDINGS.md):** NO — Helix OTA is NOT OpenDesign-refined. TWO React frontends exist (clients/ota-manager shadcn/Tauri; dashboard hand-rolled+Playwright/axe); OpenDesign (§11.4.162) not installed in either; host-render visual-proof (§11.4.170) unmet in both. The project CLAUDE.md “§11.4.162 latent” note is stale. - Validation: pre\_build PASS, inheritance PASS, regression 9/9, meta 6/6 bluff-proof, device 10× deterministic; independent review GO.

**Three parallel streams COMPLETE (§11.4.147 registry: A/B/C all complete), evidence committed:** - **A — new-tech engine adoption** → docs/research/new\_tech\_adoption\_20260709/ADOPTION\_PLAN.md. Finding (§11.4.6/§11.4.124): server has NO LLM request path today → wiring either engine now = dead code. **token\_optimizer** = adopt-as-available (first use: deterministic log\_triage over the §11.4.128 recording corpus at first AI feature); **session\_orchestrator** = needs-review (its claim+scheduler are the §11.4.176/§11.4.119 single-owner device-claim primitive — a prototype target for parallel device testing). Both build+test GREEN. Neither operator-blocked. §11.4.141 token-efficiency wins are separate Claude-Code config. - **B — server health** → docs/research/server\_health\_20260709/REMIEDIATION.md. go build/vet/gofmt/test all GREEN, 15/15 pkgs (forced -count=1 non-cached). **No fix required** (zero-finding tree; correctly changed nothing per §11.4.102). Tracked gap unchanged: internal/api/manager-dist has no \_test.go. - **C — OpenDesign + §11.4.170 harness** → docs/research/ui\_ux\_opendesign\_audit\_20260709/OPENDSIGN\_PLAN.md. Recommends **clients/ota-manager canonical** (already has the token/dark-mode arch OpenDesign feeds; dashboard would be a rebuild; dashboard’s Playwright/axe e2e is portable). §11.4.170 harness spec: Storybook 8 + addon-themes × Playwright toHaveScreenshot() golden-diff + Tesseract-OCR layout oracle, self-validated golden fixtures (§11.4.107(10)).

**BOTH OPERATOR DECISIONS RESOLVED (2026-07-09):** 1. **Canonical frontend** → **KEEP + IMPROVE BOTH** (clients/ota-manager + dashboard). Nothing retired (§11.4.122 satisfied). OpenDesign adopted across both. 2. **OpenDesign = nexu-io/open-design** (git@github.com:nexu-io/open-design.git) **CONFIRMED** by operator.

**OpenDesign ground truth (W1 evidence, docs/research/opendesign\_integration\_20260709/INTEGRATION\_GROUND\_TRUTH.md):** it is NOT a CSS-token npm package — it is a local-first design PRODUCT (Electron + od daemon + Next.js UI) exposing itself to coding agents over stdio-MCP + od CLI, shipping 153 design-system token packages as static files. **Adoption = combination:** (author-time) build daemon +

wire its MCP plugin into Claude Code; (build-time, what ships) vendor a design system's static tokens.css (~56 :root custom props) + tailwind-v4.css (Tailwind-v4 @theme, matches ota-manager's Tailwind 4), light+dark first-class. Do NOT consume @open-design/components (private, React-18.3.1-pinned ⇒ conflicts with ota-manager React 19). §11.4.74 path: author a design-systems/helix-ota/ brand package. Verdict NEEDS-REVIEW (heavy product; od daemon = network service ⇒ rootless/containerized §11.4.161/§11.4.173; keep React-18 components out).

**§11.4.170 host-render harnesses LANDED on BOTH frontends (W2/W3, independent-review GO each):** - **ota-manager** (docs/qa/20260709-ota-manager-hostrender/, harness clients/ota-manager/visual/): LoginPage × {light,dark} rendered host-side (Playwright+Vite+Tailwind4+pixelmatch+**Tesseract OCR**); dual oracle image-diff good 0% / bad flagged both themes + layout oracle flags collapsed-submit; self-validation image\_diff\_analyzer\_sound+layout\_analyzer\_sound=true (golden-good/golden-bad §11.4.107(10)); reviewer re-ran byte-identical, GO. - **dashboard** (docs/qa/20260709-dashboard-hostrender/, harness dashboard/hostrender/): Login host-rendered (vite-only, backend-independent), pixelmatch good 0% / bad 13% + layout oracle + committed golden REJECTs mutation; reviewer's adversarial negative-control (mutation-disabled → self-checks correctly FAIL) proves non-rubber-stamp, GO. Existing suite baselined: vitest 93/93, Playwright 22/1-pre-existing-flake, axe 0 critical/serious.

**NEW TRACKED BUG (real, surfaced by W2 host-render proof — §11.4.170 working as designed):** ota-manager's ui-store theme value is **never wired to a DOM .dark/.light class** (ui-store.ts:27-32 mutates zustand only; topbar.tsx swaps just the Sun/Moon icon; no document.documentElement.classList writer in src/) → the toggle changes only the icon, :root(dark) always applies. To be FIXED by the OpenDesign light/dark wiring.

**§11.4.170 ledger / Minor follow-ups (tracked, non-blocking):** (a) OCR oracle not yet mutation-validated on ota-manager — add an OCR golden-bad; (b) coverage is 1 screen/state on each frontend — remaining screen×state×{light,dark} matrix owed; (c) dashboard has NO dark mode in code (light-only proven, honest gap); (d) pre-existing ota-manager tsconfig TS6306/6310 (tsconfig.node.json lacks composite:true) — pre-existing, unrelated.

**NEXT (feature/opendesign-adoption, §11.4.167):** author design-systems/helix-ota/ brand tokens (light+dark, brand colors) → vendor tokens.css+tailwind-v4.css into BOTH frontends → **wire the ota-manager theme toggle to the DOM class + add dashboard dark mode** → prove light+dark via the two harnesses → expand the screen×state matrix + add OCR golden-bad. Also queued (non-UI): TEST-**COVERAGE** Phase 1 remainder (2026-06-23 signed-pipeline + challenges-bank still UNTRACKED, needs re-validation §11.4.108); commit\_all.sh --paths deletion-paths spec fix. **OPERATOR: rotate the posted password (§11.4.10).**

## **Latest session (2026-06-23) — Comprehensive test-coverage + anti-bluff program (Phase 0 DONE + Phase 1 mostly done)**

Continuation of the operator mandate (genuine 100% real coverage by every test type + Challenges + HelixQA vs the real running system, anti-bluff everywhere, rock-solid physical proof, no false positives). Honest framing (§11.4.6): multi-phase program — docs/research/MASTER\_TEST\_COVERAGE\_PLAN\_20260623.md (Rev 2) holds the plan + the **live coverage ledger**. This round drove **Phase 1 risk-descending (§11.4.132)** deep; all items below have real captured evidence committed + 4-remote synced (HEAD 21338527), except the two explicitly marked IN FLIGHT:

- **Integration (pgx)** — store **85.5%** / rollout **83.1%** MEASURED via real -tags integration podman+Postgres run (docs/qa/20260623-postgres-integration/).
- **F-ANTIBLUFF-LIB** — tests/lib/anti\_bluff.sh  
ab\_pass\_with\_evidence per-PASS §11.4.69 helper, mutation-proven self-test + standing guard (docs/qa/20260623-antibluff-lib/).
- **F-CLUSTER real-system boot** — server/deploy/system.compose.yml + tests/lib/boot\_real\_system.sh boot the REAL ota-server + real Postgres → /readyz → 200 + DB-backed admin JWT login (docs/qa/20260623-real-system-boot/); two §11.4.102 prod fixes (startup retry-with-backoff, harness clean-slate).
- **e2e vs REAL DB** — challenge\_operational **39/39 PASS** against live Postgres (DB-proof: 14 audit\_logs rows, real end-of-run delete; docs/qa/20260623-e2e-live-system/).
- **Security trust-boundary 4/4 vs live** — runtime proof the **request-supplied artifact pubkey is IGNORED** (server-config key only); throwaway test secrets redacted (§11.4.10, commits 84f32a66/1d71e182). Evidence docs/qa/20260623-trust-boundary-live/.

- **Go fuzz** — 4 targets, **10.8M execs / 0 crashers**, property-based (round-trip / antisymmetry / hex-oracle); docs/qa/20260623-fuzz/.
- **HTTP load** — p99 **14.32ms** @ 5,540 req/s, zero 5xx; docs/qa/20260623-http-load-live/.
- **Chaos (§11.4.85) 4/4 survive+recover** — postgres-kill → reconnect (validates main.go retry fix, no corruption), malformed/oversized → 400, 12-race idempotent register → exactly-one-device, 400-conn churn → recovers; docs/qa/20260623-chaos-live/.
- **Benchmark** — benchstat baseline registry + negation-proven guard (7 benches; thresholds calibrated on measured variance); docs/benchmarks/ + docs/qa/20260623-benchmarks/.
- **Android JaCoCo** — wired both bricks: ota-update-engine-bridge **LINE 100%** (113/113), ota-android-agent **LINE 91.18%** (json-codec branch 55.6% → **100%/100%** via 54 new tests); docs/qa/20260623-android-jacoco/ + docs/qa/20260623-json-coverage/.
- **F-METAGATES + propagation-gate batch** — meta-test framework wired into pre-build; **ALL 14 propagation gates now §1.1-paired** (data-driven from pre\_build source → new gates auto-covered); §11.4.166 semgrep fail-open FIXED; one real §11.4.50 flake found+fixed (10/10 det). docs/qa/20260623-metagates/ + docs/qa/20260623-propagation-metagates/.
- **Independent review (§11.4.165)** — the batch passed an independent verifier that **found + fixed a tautological restore-integrity bluff** in tests/meta/lib\_metatest.sh.
- **IN FLIGHT (not yet committed, §11.4.6)**: signed-pipeline-vs-live (tests/e2e/pipeline\_signed\_live.sh — a local run is **15/15 PASS**: signed upload → release → deploy → rollout → device-poll-receives-signed-update vs live system, but docs/qa/20260623-signed-pipeline-live/ is UNTRACKED) and the Challenges-bank dry-run (docs/qa/20260623-challenges-bank/, UNTRACKED). The conductor lands these.

**NEXT (Phase 1 remainder + Phase 2, risk-ordered)**: (1) commit the in-flight signed-pipeline-vs-live + challenges-bank evidence; (2) security **saturation / DDoS** (rate-limit flood vs live); (3) **on-device agent A/B** (drive the two JVM bricks on the live Cuttlefish cvd, or RK3588 when reachable); (4) **Phase 2** — wire the Go cmd/helixqa orchestrator + new OTA Challenges (bad-sig-rejected, request-key-ignored, telemetry-schema, rollout-staged+halt, chaos) against the live system; (5) functional §11.4.165 media/vision gate (the §11.4.163 pipeline) beyond the now-bluff-proof grep-gate layer. Suites: regression 8/8 + meta 5/5 GREEN. cvd still live on nezha. **OPERATOR: rotate the posted password (§11.4.10).**

## Prior session (2026-06-23) — Comprehensive test-coverage + anti-bluff program (Phase 0 + Phase 1 start)

Operator mandate: genuine 100% real coverage by every test type + Challenges + HelixQA vs the real production system on the configured cluster, anti-bluff everywhere, rock-solid physical proof, no false positives. Honest framing (§11.4.6): this is a multi-phase program — docs/research/MASTER\_TEST\_COVERAGE\_PLAN\_20260623.md holds the plan + a **live coverage ledger** (updated as each item lands). Landed this round, all real captured evidence, committed + 4-remote synced:

- **Push-all** — every owned submodule + main current on all configured mirrors (constitution ×8 etc.).
- **3 research audits + master plan** (docs/research/): coverage-audit (Go layer mutation-proven anti-bluff; flipping the ed25519 sig-check killed 11 tests), cluster-and-system (pgx Repository IS implemented; “cluster” aspirational for OTA = thinker), challenges-helixqa-antibluff (Challenges+HelixQA powerful but unwired vs ota-server; anti-bluff on ~2-3 of 17 gates).
- **Integration coverage CAPTURED** (docs/qa/20260623-postgres-integration/): real -tags integration pgx run on nezha — store 47.9% → 85.5%, rollout 28.2% → 83.1%.
- **F-ANTIBLUFF-LIB** (tests/lib/anti\_bluff.sh, docs/qa/20260623-antibluff-lib/): the §11.4.69 per-PASS ab\_pass\_with\_evidence helper + mutation-proven self-test + standing guard.
- **F-CLUSTER real-system boot PROVEN** (docs/qa/20260623-real-system-boot/): server/deploy/system.compose.yml + tests/lib/boot\_real\_system.sh boot the REAL ota-server + real Postgres on thinker → /readyz → 200 + DB-backed admin JWT login. Two §11.4.102 production fixes: server startup retry-with-backoff (was log.Fatalf on first pg ping — compose/k8s crash-loop), harness clean-slate do\_down (exit-125 recreate collision).
- **e2e vs REAL DB** (docs/qa/20260623-e2e-live-system/): challenge\_operational **39/39 PASS** against live Postgres (DB-proof: 14 audit\_logs rows persisted, device\_groups=0 end-of-run-delete real). Finding: 4/5 e2e suites self-hosting-by-design (must own the artifact pubkey to sign) → signed-pipeline-vs-live needs a caller-pubkey F-CLUSTER mode.
- **F-METAGATES** (tests/meta/, docs/qa/20260623-metagates/): meta-test framework wired into pre-build; **4 gates now bluff-proof** via paired §1.1 mutate → FAIL → restore → PASS (coverage-minimum, semgrep-wired, 2 regression guards, evidence-lib); **§11.4.166**

**semgrep fail-open FIXED** (semgrep on PATH → gate passes); a real flake found+fixed (§11.4.50, 10/10 det).

**NEXT (Phase 1 risk-ordered, all documented in the master plan):**  
signed-pipeline-vs-live (caller-pubkey F-CLUSTER mode) · security trust-boundary negatives + go test -fuzz · HTTP load/latency p50/p95/p99 (wire tools/loadtest) · Android JaCoCo + on-device A/B · remaining ~14 anchor-greps → functional/paired gates · Phase 2  
Challenges+HelixQA Go orchestrator vs live system. Suites: regression 8/8 + meta 4/4 GREEN. cvd still live on nezha. **OPERATOR: rotate the posted password (§11.4.10).**

## **Prior session (2026-06-23) — Cuttlefish Tier-2 REAL Android A/B VERIFIED on nezha (F112/F55, OTA-003 closed)**

**TERMINAL GOAL MET (honest §11.4.6 — real captured evidence).** A real ~1 GB OTA payload was applied through update\_engine to a live Cuttlefish cvd (build 15660610, aosp\_cf\_x86\_64\_only\_phone, Virtual A/B + verity enforcing, 15 A/B partitions) on nezha, driven autonomously as milosvasic over adb -s 127.0.0.1:6520 (no host sudo for the A/B flow):  
onPayloadApplicationComplete(kSuccess) →  
UPDATE\_STATUS\_UPDATED\_NEED\_REBOOT (115 s) → reboot slot flip \_a→\_b (VAB merge merging→none, \_b marked successful) → forced-bad slot \_a (bootctl set-slot-as-unbootable + bounded 256 KB inactive-slot boot\_a write, §11.4.133) rejected → device booted known-good \_b. The OTA payload was obtained with NO credentials (androidbuildinternal pre-signed GCS URL storage.googleapis.com, 1003473429 B, md5 d90870a9a6eece3868520d7fd3f098c — size+md5 verified before apply).

- **Evidence:** docs/qa/20260623-cuttlefish-tier2-ab/REPORT.md (+ apply\_full.log, slot\_flip.log, rollback.log, corrupt\_dd.txt, ab\_facts.txt, ...) — read-the-screen verified per §11.4.158 (REPORT §7).
- **Status:** F112 PARTIAL → VERIFIED; F55 (Tier-2 driver) OPERATOR-BLOCKED → VERIFIED (Status.md rev 21, Status\_Summary rev 13; VERIFIED 25 → 27, PARTIAL 4 → 3, OPERATOR-BLOCKED 2 → 1).
- **Validator:** tests/emulator/tier2-cuttlefish\_ab.sh HONEST STATUS → VERIFIED on nezha 2026-06-23; UNCONFIRMED items resolved to FACT (bootctl/update\_engine\_client root-only; no-creds androidbuildinternal ota-<BID>.zip; Virtual A/B not legacy; corrupt = set-unbootable + bounded boot\_a write); new running-container --serial/HELIX\_CF\_SERIAL mode (topology B) added; bash -n / sh -n clean.



- **Regression guard:** tests/regression/guard\_cuttlefish\_ab\_proven.sh GREEN (§11.4.135; asserts slot\_flip/rollback/kSuccess evidence + validator VERIFIED header, RED on stripped proof).
- **Full journey (provenance):** curl-download-fail → resumable wget-c recovery → 27.6 GB single-stage image → slim 1.11 GB prebuilt-deb path (containers 54aa9b2) → operator privileged launch (cf-launch.sh) → cvd booted → this A/B PASS. **cvd left running on nezha.**
- **Honest boundary (§11.4.3/§11.4.112/§11.4.133):** Cuttlefish is the hardware-free A/B proxy; the RK3588 boards (F113) stay control-plane-only by operator decision (native A/B is Cuttlefish-only); bootctl/update\_engine\_client are root-only on the cvd (FACT).

## Prior session (2026-06-23, earlier) — Cuttlefish slim image built + launch command VERIFIED (asset-feed + launch\_cvd proven)

Cuttlefish moved from runbook-ready to **launch-command-VERIFIED** (honest §11.4.6 — still NOT a real-A/B PASS):

1. **Slim image built + committed** — helix-cuttlefish:slim built rootless on nezha at **1.11 GB** (vs 27.6 GB single-stage from-source) via the upstream runner-prod **prebuilt-.deb** path (cuttlefish-base/cuttlefish-user **1.54.1** from us-apt.pkg.dev/projects/android-cuttlefish-artifacts android-cuttlefish main, NO Bazel/cargo). cvd version 1.54.1 executes. containers submodule 54aa9b2; parent pointer **659c2326**. Saved to /tmp/cf-slim.tar (1.03 GiB) for rootless → rootful load.
2. **Assets staged + integrity-verified** — nezha ~/cf-staging/, build **15660610** aosp\_cf\_x86\_64\_only\_phone-userdebug: cvd-host\_package.tar.gz (898828370 B, gzip-valid) + img.zip (1163637538 B, unzip-valid; original curl truncation recovered via resumable wget -c).
3. **Runtime model FACT (§11.4.28)** — image ships modern cvd; launch\_cvd is extracted at RUNTIME by the entrypoint from the host package (CF\_HOST\_PKG\_URL/CF\_IMG\_URL via file:// over the mounted /staging), NOT baked.
4. **PRE-VERIFY PROOF (§4.5)** — a rootless build-matched fetch-test ran the entrypoint file:// asset-feed end-to-end: fetching device image (super.img + boot/init\_boot/vbmeta extracted) → fetching host package (./bin/launch\_cvd present) → launching cvd via ./bin/launch\_cvd → **launch\_cvd RAN**, assembled the cvd-1 config, Launcher Build ID: 15660610; then EXPECTED rootless

VIRTUAL\_DEVICE\_BOOT\_FAILED run\_cvd returned 10 (no /dev/kvm/bridge). Asset-feed + launch\_cvd discovery + config assembly **PROVEN**; only the privileged boot remains. Evidence: docs/qa/20260623-cuttlefish-launch-verified/REPORT.md.

5. **VERIFIED operator privileged launch** (runbook §2.3): sudo modprobe vhost\_vsock → sudo podman load -i /tmp/cf-slim.tar → sudo podman run -d --name cuttlefish --privileged --network host --device /dev/kvm ...vhost-vsock ...vhost-net ...vsock ...net/tun -v /home/milosvasic/cf-staging:/staging:ro -e CF\_HOST\_PKG\_URL=file:///staging/cvd-host\_package.tar.gz -e CF\_IMG\_URL=file:///staging/img.zip helix-cuttlefish:slim → sudo podman logs -f cuttlefish. Rootless → rootful gap closed by the save|load step (§11.4.161 exception — privileged run is rootful).

**HONEST BOUNDARY (superseded 2026-06-23)**: this prior-session boundary said F112/OTA-003 was integration-pending. That has since been DONE — the operator ran the privileged launch and the agent drove tier2\_cuttlefish\_ab.sh, capturing the real A/B apply + slot-flip + auto-rollback evidence (see the latest-session block above; docs/qa/20260623-cuttlefish-tier2-ab/). F112/F55 are VERIFIED.

**Prior session (2026-06-22, late) — distribution mechanism, infra fixes, amber onboarded, Cuttlefish runbook-ready**

**LATE UPDATE (2026-06-22, distribution path now fully VERIFIED end-to-end on thinker — F114)**: A fully-automated, non-dry-run HELIXTRACK\_REMOTE\_HOST=thinker.local bash scripts/distribute\_stack.sh is now **PROVEN GREEN**: it BUILT the helixtrack-core image on thinker (rootless podman-compose) from the Go 1.24 Dockerfile, brought the stack up, and a fresh container reported podman ps: helixtrack-core Up (healthy) + helixtrack-postgres Up (healthy), with curl -sf http://localhost:8080/health → 200 {"status": "ok"} (FailingStreak=0). Evidence docs/qa/20260622-222645-distribute-thinker-FULLY-GREEN/. The fix chain that closed the Rev-19 blockers: distribute\_stack.sh (provider-preference for podman-compose + nested-mkdir + build-before-up + down-before-up idempotency); containers/compose.helixtrack.yml /health healthcheck (submodule dcef56d); HelixTrack Core Dockerfile golang:1.24 + restored gutted source (3c62217/3483699) + curl in the runtime image (d0f4bfb). **F114 is now VERIFIED** (Status.md rev 20, Status\_Summary rev 12; VERIFIED

24 → 25, PARTIAL 5 → 4). **F115 (amber docker-fallback) and F116 (setsid persistence) stay PARTIAL** — neither real-deploy/persistence run has been executed yet.

Docs + infra consolidation (honest §11.4.6 — no new PASS claimed):

- 1. Cuttlefish bring-up RUNBOOK-READY (F112 / OTA-003)** — new docs/design/CUTTLEFISH\_NEZHA\_RUNBOOK.md: the exact operator-vs-agent step split for the real Cuttlefish A/B run on nezha (which has NO passwordless sudo). **Operator** runs the 3 privileged steps — (§2.1) verify/load vhost\_vsock/vhost\_net + create /dev/vsock if absent, (§2.2) one-time group membership, (§2.3) the sudo podman run --privileged --network host --device /dev/kvm ...vhost-vsock ...vhost-net ...vsock ...net/tun -v ~/cf-staging:/staging cuttlefish:latest. **Agent** drives the rest — build the image (rootless), extract the staged assets (~/.cf-staging/cvd-host\_package.tar.gz 898 MB + img.zip 1.16 GB), launch\_cvd --daemon, the tier2\_cuttlefish\_ab.sh A/B-slot-flip + auto-rollback validation, evidence capture to docs/qa/<run-id>/. Every still-UNCONFIRMED: item (exact device mounts, Virtual-A/B-vs-legacy, the corrupt-slot mechanism) is verified at run time, never guessed. **HONEST BOUNDARY: this is a runbook to EXECUTE — NOT a real-A/B PASS.** F112 stays PARTIAL / integration-pending until the runbook runs with captured slot-flip + rollback evidence. Built under the §11.4.161 documented exception (CUTTLEFISH\_ROOTFUL\_EXCEPTION.md).
- 2. Distribution mechanism + amber onboarded (F114/F115)** — distribute\_stack.sh (helix layer, §11.4.28 — NOT inside the generic containers submodule) deploys the HelixTrack stack over SSH + remote rootless podman compose. thinker.local = LIVE rootless-podman target; amber.local onboarded 2026-06-22 (SSH key installed + docker present) with a **§11.4.161 operator-authorized docker fallback** (HELIX\_ALLOW\_DOCKER\_FALLBACK=1, default-OFF rootless-or-nothing, §11.4.112 documented constraint = no rootless podman on amber yet); nezha.local is read/import-only. Companion doc docs/scripts/distribute\_stack.md updated (Rev 2) with the docker-fallback section.
- 3. Remote-emulator full-detachment fix (F116, §11.4.144)** — scripts/boot\_android\_emulator.sh remote launch wrapped in setsid nohup ... </dev/null >log 2>&1 & (own session + process group) so an interrupted launching SSH session no longer kills the remote emulator (closes the known robustness gap noted in Rev 5). Companion doc docs/scripts/boot\_android\_emulator.md Rev 2. Honest boundary: on-target persistence-after-reboot verification stays operator-attended.

4. **commit\_all cascade-push fix (F117)** — `commit_all.sh/push_all.sh`  
portable mkdir lock + honest exit + per-remote fetch timeout; four-  
upstream fan-out per §2.1/§11.4.88.

Docs: 4 new Status feature rows (F114–F117, all VERIFIED), Status.md  
Rev 18 + Status\_Summary.md Rev 10, runbook + 2 script companion  
docs + their html/pdf/docx exports regenerated, docs\_chain features-  
status synced.

**Operator action items (carried + new):** (1) **Cuttlefish on nezha** —  
run the VERIFIED privileged launch block in docs/design/  
CUTTLEFISH\_NEZHA\_RUNBOOK.md §2.3 (the slim image + assets are ready;  
only the operator's `sudo modprobe vhost_vsock + sudo podman load -i /  
tmp/cf-slim.tar + sudo podman run --privileged ...` remains) to unblock  
the real Android A/B run — the agent then drives the A/B validation; (2)  
**amber** — install rootless podman to retire the §11.4.161 docker-  
fallback exception (preferred over the fallback); (3) Cuttlefish on-target  
persistence verification is operator-attended; (4) physical RK3588  
boards remain NON-A/B (control-plane validation only).

## **Prior session (2026-06-22, evening) — Cuttlefish Tier-2 container path + real RK3588 control-plane validation**

Three new real capabilities landed (honest §11.4.6 — boundaries  
stated, not overclaimed):

1. **Cuttlefish Tier-2 containerized path (F112)** — new pkg/  
cuttlefish cvd-lifecycle wrapper in the containers submodule  
(cuttlefish.go / accel.go / cleanup.go / health.go / types.go /  
entrypoint.sh / Containerfile). `go test -race ./pkg/cuttlefish/ =`  
`**30 PASS`
  - 1 honest topology SKIP\*\* (no Linux+KVM on this macOS host).  
Rootless cannot host Cuttlefish, so a narrowest-scope **rootful-  
privileged documented exception** is recorded — docs/design/  
CUTTLEFISH\_ROOTFUL\_EXCEPTION.md (§11.4.161 documented  
exception via §11.4.112; image build + artifact fetch stay  
rootless, only `launch_cvd` is privileged). **HONEST BOUNDARY:**  
**the container path is BUILT + unit-tested — it is NOT yet a  
real-A/B PASS.** The real end-to-end Android `update_engine/AVB/  
dm-verity A/B` run is **integration-pending** on nezha Linux+KVM  
(assets staging; privileged `launch_cvd` operator-gated). Native A/  
B fidelity is this path's job once provisioned.
2. **Real RK3588 hardware control-plane validation (F113)** —  
evidence docs/qa/20260622-rk3588-controlplane/REPORT.md. Server  
cross-built linux/amd64, run **rootless** on nezha (uid 1000). **Device**

**B** (Ethernet, serial 1acdceab90248933) **PASS**: board-originated GET / healthz 200, GET /api/v1/client/update 204 (no active deployment — correct), POST /api/v1/client/telemetry 202, and sink-side GET / devices/by-hardware/1acdceab90248933 → update\_state=success (the board’s own telemetry mutated server state). **Device A** (Wi-Fi, 19bbb528a1dbbc4d) honest topology **SKIP** — VPN tun1 full-tunnel / Wi-Fi AP isolation (busybox nc to both :18080 and :22 time out → blocked network path, captured root cause, NOT a Helix defect). **HONEST BOUNDARY: both boards are NON-A/B** (single-slot, no update\_engine) → this validates the control plane on real hardware, NOT native A/B apply. **ZERO device state changes** (§11.4.122/ §11.4.133).

3. **Distribution repoint** — container distribution targets now thinker.local (live) + amber.local (SSH-key-pending); nezha.local is read/import-only.

**Operator action items:** (1) **amber.local** — install the SSH key (ssh-copy-id to amber) to bring it live as a distribution target; (2) **Cuttlefish on nezha** — the privileged launch\_cvd needs operator sudo (+ reboot + ~30 GB fetch\_cvd) to unblock the real Android A/B run (F112 integration-pending); (3) **assets staging in progress** for the Cuttlefish Tier-2 run; (4) the **physical RK3588 boards are NON-A/B** (single-slot) — real on-device A/B fidelity is the Cuttlefish path’s job, not these boards (control-plane validation only on them).

## **Prior session (2026-06-22, daytime) — “do everything workable” sweep**

All autonomous items GREEN with real captured evidence (no bluffs): - Test sweep GREEN (server 11/11 + 4 Go submodules 3/3, -race 0); **gofmt fixed** (13 files); pre-build gates PASS; docs\_chain in-sync. - **Local QEMU A/B OTA 6/6 GREEN** (smoke/boot/slot-switch/rollback/OTA-apply) — evidence docs/qa/20260622T07\*. Fixed a real §11.4.115 RED-mode polarity isolation bug in ab\_rauc\_verity.sh (RED×2+GREEN×2 deterministic). - **10 server-feature recordings regenerated**, vision-verified (203 HTTP assertions) — docs/qa/20260622-server-recordings-regen/; Status.md §11.4.153 reconciled to durable paths. - **§11.4.166 Semgrep**: propagated §11.4.160-166 into CLAUDE/AGENTS/GEMINI; tokenless local scan wired into commit\_all.sh; constitution pin → 09d8940 (adds docs/semgrep/TOKEN\_SETUP.md); 7 findings triaged → **0 remaining** (TLS MinVersion fix + cited suppressions). - Submodule push parity closed (ota-update-engine-bridge, ota-android-agent); GitFlic “blocker” disproven; §11.4.30 transient qa-results/

untracked+ignored (232 files). - nezha AVD boot proven; real Android A/B = honest **operator-attended SKIP** (Cuttlefish unprovisioned — needs sudo+reboot+fetch\_cvd).

**Operator action items:** (1) Semgrep SEMGREP\_APP\_TOKEN — follow constitution/docs/semgrep/TOKEN\_SETUP.md (semgrep login) to silence the MCP hook (optional; tokenless gate already compliant); (2) provision Cuttlefish on nezha to unblock real Android A/B; (3) RK3588 board for OTA-004/F55/F56.

**Known robustness gap (tracked, NOT yet fixed — §11.4.6/§11.4.123):** scripts/boot\_android\_emulator.sh — the remote qemu/AVD on nezha stays tied to the launching SSH session and exits gracefully if that session ends mid-boot-wait (the nohup does not fully detach the remote process; the final SSH-tunnel/attestation step isn’t reached on interrupt). Boot itself is PROVEN (emulator-5554, API 36, boot\_completed=1, evidence qa-results/20260622T071848Z-nezha-android-ab/). Fix candidate: wrap the remote launch in setsid nohup ... </dev/null >log 2>&1 & (or a systemd-run --user --scope on nezha) for true detachment — deferred because on-target persistence verification (re-boot + interrupt test, leaves remote state) is operator-attended; a blind change to the working boot path is forbidden per §11.4.1 without rock-solid proof.

Active items

| ID      | Title                                                                               | Status                                                                                                | Type |
|---------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------|
| OTA-003 | Emulator Tier-2 — real Android A/B (update_engine/AVB/dm-verity auto-rollback)      | Completed (→ Fixed.md) — VERIFIED on nezha 2026-06-23, evidence docs/qa/20260623-cuttlefish-tier2-ab/ | Task |
| OTA-004 | Emulator Tier-3 — real RK3588 / Orange Pi 5 Max vendor HAL, U-Boot slot-switch, dm- | Operator-blocked                                                                                      | Task |

| ID | Title                     | Status | Type |
|----|---------------------------|--------|------|
|    | verity on real partitions |        |      |

## Recently closed items

| ID      | Title                                                 | Status      |
|---------|-------------------------------------------------------|-------------|
| OTA-021 | HelixTrack<br>bidirectional sync<br>verification      | Completed   |
| OTA-020 | Database migration<br>test                            | Fixed       |
| OTA-019 | Build resource stats<br>tracker                       | Completed   |
| OTA-018 | ApplyPort CLI + slot<br>manager + Ed25519<br>verifier | Implemented |
| OTA-017 | U-Boot corrupt-slot<br>auto-rollback via<br>bootcount | Implemented |
| OTA-016 | RAUC dd-apply to<br>inactive slot with dm-<br>verity  | Implemented |
| OTA-015 | A/B slot switch via U-<br>Boot BOOT_ORDER             | Implemented |

Full issue details: [docs/Issues.md](#) · [docs/Fixed.md](#) · [docs/Issues\\_Summary.md](#) · [docs/Fixed\\_Summary.md](#).

## 2. Server tests — all GREEN

All 13 testable packages pass (11 internal packages + chaos + stress suites):

```
ok  internal/api
ok  internal/config
ok  internal/device
ok  internal/deviceemu
ok  internal/fabric
ok  internal/health
```

```
ok internal/rollout
ok internal/store
ok internal/transport
ok tests/chaos
ok tests/stress
```

No-test-file packages (build-only): cmd/applyport, cmd/ota-device-emu, cmd/ota-server, tools/loadtest.

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### 3. Emulator status

| Property          | Value                                                                    |
|-------------------|--------------------------------------------------------------------------|
| Host              | nezha.local — Linux x86_64, 62 GB RAM, KVM                               |
| Image             | Android API 36 (Android 16) — CZ_API36_Phone                             |
| Transport         | ADB over SSH tunnel (reachable at emulator-5554 from nezha)              |
| ADB local         | No Android devices connected locally (emulator is remote via SSH tunnel) |
| LD_LIBRARY_PATH   | /home/milosvasic/.local/lib                                              |
| Cuttlefish Tier-2 | Pending AOSP guest images                                                |

The HelixTrack API is accessible from the emulator via SSH tunnel. The CZ\_API36\_Phone image provides the standard Android 16 AOSP stack on an x86\_64 KVM host.

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### 4. What's running

- **Main stream:** Emulator Tier-2 validation (remote on nezha.local)
  - **Background agents:** None currently dispatched
  - **Recording directory:** \$HOME/Downloads per §11.4.158(D) (default; no project-level override)
-



## 5. Next actions (priority-ordered)

1. **OTA-003 — DONE (VERIFIED on nezha 2026-06-23).** Real Android A/B (update\_engine apply → slot flip \_a→\_b → auto-rollback) proven on a live Cuttlefish cvd; evidence docs/qa/20260623-cuttlefish-tier2-ab/REPORT.md; §11.4.135 guard GREEN. Migrate the Issues.md entry → Fixed.md (conductor) and confirm exports.
  2. **OTA-004 — Hardware unblock.** When a physical RK3588 / Orange Pi 5 Max board becomes reachable over ADB/SSH, flash and validate Tier-3 (vendor HAL, U-Boot slot-switch, real-partition dm-verity). Boards stay control-plane-only by operator decision (§11.4.133) until then.
  3. **Feature-coverage video recording.** Produce §11.4.153 mandatory per-feature real-use videos confirming every server endpoint, every emulator tier, and every submodule works — with §11.4.159 window-scoped MP4 capture + §11.4.160 vision verification.
  4. **Standing regression-guard suite (§11.4.135).** Ensure every closed OTA-NNN item has its §11.4.115 polarity-switch regression test registered in the suite.
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## 6. Binding constraints

- **Anti-bluff (§11.4 / §107):** Every PASS carries captured physical evidence per §11.4.5 / §11.4.69 / §11.4.107. Metadata-only / config-only / absence-of-error / grep-without-runtime PASS are all forbidden.
- **No-force-push (§11.4.113):** git push --force, --force-with-lease, +<ref> are STRICTLY FORBIDDEN. Always merge onto latest main and push fast-forward-only.
- **Commit via commit\_all.sh:** All changes use the project's canonical commit wrapper. Direct git add/git commit/git push are never used.
- **Four-layer fix-verification (§11.4.108):** SOURCE → ARTIFACT → RUNTIME-ON-CLEAN-TARGET → USER-VISIBLE. Runtime signature is the definition of done.
- **Independent review (§11.4.142 / §11.4.125):** Every change passes an independent code-review agent before build. Review iterates to zero-finding GO per §11.4.134.
- **Multi-upstream push (§2.1):** Every commit fans out to all four upstreams (GitHub + GitLab + GitFlic + GitVerse).
- **Rootless containers (§11.4.161):** All containerized workloads use Podman in rootless mode via vasic-digital/containers submodule.

- **No remote CI (§11.4.156):** All CI/CD automation is DISABLED. Enforcement is through local git hooks + `pre_build_verification.sh`.
- 

## 7. Feature tracking

Feature inventory and per-row status: `docs/features/Status.md` (Rev 18, 2026-06-22). Summary companion: `docs/features/Status_Summary.md`.

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## 8. Fresh session start

```
git fetch --all --prune --tags
```

Then read this file (`docs/CONTINUATION.md`) and `docs/Issues.md` for the full active-item context. **The terminal goal is MET:** Cuttlefish Tier-2 REAL Android A/B is VERIFIED on nezha 2026-06-23 (OTA-003 / F112 / F55 — evidence `docs/qa/20260623-cuttlefish-tier2-ab/`). The remaining open item is OTA-004 (Tier-3 physical RK3588), which is operator-blocked on hardware (boards stay control-plane-only by operator decision, §11.4.133). The cvd is left running on nezha.